

AGORA

One Sentence, One Living World

Generating Persistent Worlds for Human–AI Social Interaction

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Abstract. A generated social world needs a connection between its description and what its inhabitants can actually change. We present Agora, a mechanism for compiling a short premise into a persistent rendered environment for human–AI interaction. A shared world representation binds roles, places, possessions, knowledge, and visual assets to stable entity identities; dependency-directed validation and repair keep these components consistent. A generated action repertoire makes situated activities discoverable, while an open-proposal channel expresses intentions beyond that repertoire. The World Coordinator checks structured effects against the generated state before committing changes that become available to subsequent actors. The central contribution is this coupling of world construction, situated action, and persistent consequences. We examine its operation through construction checks, execution traces, and an existing visual study with ten eligible respondents. Validators detect all 58 seeded faults; five local repairs preserve unaffected generation stages. Five generated worlds exercise both action pathways. Longer scripted traces expose committed inventory and location changes, rejected effects, and the distinction between an executed action and useful task progress. Supplementary planner-input and action-space ablations document implementation behavior with replayable records. These studies establish the implemented connection from generated world structure to checkable state transitions; human collaboration and behavioral fidelity remain separate evaluation questions.



Figure 1. Five generated settings share a world schema while varying institutions, spaces, and visual identity. Panels show preserved map assets and representative character frames; they are not new experimental outputs.

1 Introduction

A generated social world needs persistent consequences. In a floating embassy where endangered languages are political assets, inhabitants must be able to access archives, trade translation rights, and respond when a flood threatens the only surviving treaty. These activities require explicit places, possessions, information, and rules behind the rendered scene.

Several research lines address parts of this problem. Procedural content generation studies representations and criteria for functional game content [1, 2]. Word2World and DreamGarden extend language-based authoring to playable games and development plans [3, 4]. Generative Agents demonstrates how local memory and encounters can produce community activities in an authored town [5], while Concordia grounds language-described actions through a Game Master [6]. These advances motivate a joint design question: how can one short premise specify both a world and the conditions under which a society can act within it?

We present Agora, a system that compiles a one-sentence premise into a persistent, visually grounded environment for humans and language agents. Its representation connects rooms, roles, institutions, possessions, private knowledge, vi-

sual assets, and action opportunities. A generated repertoire makes the world’s activities discoverable; an open-proposal channel lets an actor express a more specific intention. A World Coordinator interprets each proposal against shared constraints before its effects enter the world state. For example, proposing a treaty exchange does not by itself transfer ownership: the resources, participants, and required acceptance must support the change.

The research object is a state trajectory

$$\tau = (S_0, q_1, d_1, S_1, \dots, q_T, d_T, S_T), \quad (1)$$

where q_t is an intended action and d_t is the coordinator decision. This representation distinguishes an intention, an enacted effect, and a later response. It also makes explicit which observations concern software execution, simulated social behavior, or human experience.

Our contributions are threefold:

1. A compositional language-to-world pipeline connecting semantic, visual, spatial, and mutable social state, with dependency-directed local refinement.
2. A shared action interface for humans and agents that combines generated opportunities with coordinator-governed

proposals and persistent consequences.

3. An auditable account of how these mechanisms operate: seeded defects and local repairs test construction dependencies; interaction records expose action selection, coordinator decisions, and committed state changes. Existing visual judgments characterize the rendered artifacts, and supplementary planner diagnostics document input and action-space restrictions.

The paper asks how a premise-generated setting becomes an environment with executable consequences. Its evidence follows the mechanisms from generation contracts through action adjudication to persistent state. Model-specific artifact judgments and supplementary configuration diagnostics provide implementation context; behavioral fidelity and causal social effects remain outside the completed studies’ scope.

2 Related Work and Research Position

2.1 From Generated Content to Executable Worlds

Procedural content generation distinguishes what is represented, how it is generated, and how its fitness is evaluated [1]. PCGML further connects learned content models with repair and critique [2]. This lineage matters for Agora: decomposing a world and repairing a faulty part are established design ideas. Our use of them ties a repair to explicit dependencies among role, inventory, geometry, media, and runtime state.

Word2World converts stories into playable tile-based games and evaluates its generation stages [3]. DreamGarden recursively decomposes a user prompt into plans executed by specialized modules, with user feedback during Unreal Engine development [4]. Both are close precedents for prompt-driven, staged authoring. Agora emphasizes a reusable social-state representation: the generated institutions and objects become inputs to ownership, knowledge access, and action adjudication during play. We make no claim that prompt-based world generation or hierarchical planning is new.

Environment generation also serves embodied-agent training. ProcTHOR samples diverse interactive houses [7]; Holodeck uses language and spatial constraints to produce 3D scenes from prompts [8]. Genie instead learns action-controllable visual dynamics from video [9]. These approaches provide distinct substrates for interaction. Agora uses explicit entity identities and state transitions in a rendered 2D environment, making social and material changes directly inspectable. Visual generation and symbolic persistence address complementary aspects of world construction.

2.2 Grounded Dialogue and Shared Play

TextWorld provides generated text games with explicit state tracking and controllable task difficulty [10]. ScienceWorld evaluates whether agents can carry out scientific procedures in an interactive environment [11]. Both motivate evaluating executed consequences rather than the plausibility of a textual plan.

LIGHT is a particularly relevant predecessor: humans and models inhabit a fantasy environment, combining dialogue with actions grounded in local objects and characters [12]. Thus shared human-agent play and situated dialogue are not

unique to Agora. Our focus is generating the surrounding social and material setting from a premise, then coupling visible activities and novel proposals to the same persistent state. Human-directed events in our logs establish reachability of the player, while experience and collaboration quality require a participant study.

2.3 Persistent Communities and Social Simulation

Generative Agents combines observations, retrieval, reflection, and planning in a town of 25 agents, illustrating how remembered encounters can support coordinated activities [5]. Concordia provides a flexible simulation framework in which agents describe intentions and a Game Master grounds their effects [6]. The World Coordinator follows this separation of intention and environment response; Agora connects it to premise-generated identities, inventories, rules, and rendered spaces.

AgentSociety studies large populations and interventions in a societal environment [14]; LMAgent combines multi-modal shopping behavior with large-scale social interaction [15]. Avalon studies cooperation and confrontation under explicit game rules [16]. These systems show why trajectories and institutions matter beyond isolated dialogue quality. Our smaller studies address authoring diversity and trace inspectability, without claiming comparable scale or empirical correspondence to society.

2.4 Action, Feedback, and Feasibility

ReAct interleaves reasoning and environment actions [17]; Reflexion stores verbal feedback for subsequent attempts [18]; Voyager accumulates executable skills through environment feedback and verification [19]. These methods motivate the observation–intention–outcome cycle. SayCan explicitly connects language-level usefulness to skill feasibility [20]. Agora applies the analogous distinction to social and material effects: an intention can be intelligible while its requested transfer, construction, or status change is infeasible. Its action interface adds partner acceptance, ownership, and world-specific rules to that decision. This is a system design connection, not a claim of algorithmic equivalence or an empirical comparison with those agents.

2.5 Evaluation and Validity

SOTOPIA evaluates interaction through multiple social goals and criteria [13]. Subsequent work shows that an omniscient simulation can appear more successful than an information-asymmetric interaction [21]. This motivates our explicit observation boundary and private knowledge, but does not by itself validate their implementation. SOTOPIA- π also reports evaluator overestimation of agents trained for social interaction [22]. Together with broader LLM-judge biases [23], this motivates keeping structural checks, state-derived metrics, and the present human artifact judgments separate.

AgentBoard motivates inspecting intermediate progress rather than relying on final success alone [24]. We similarly report executed changes separately from plausible intentions, but do not claim that our mutation counts measure validated task progress. AI Agents That Matter emphasizes cost, hold-outs, and reproducibility [25]; this informs our separation of

Research line	Representative substrate	Action or generation mechanism	Relation to Agora
Word2World; DreamGarden [3, 4]	Generated game content and environments	Story expansion; hierarchical development plans	Precedents for language-driven, staged authoring
ProcTHOR; Holodeck [7, 8]	Procedural or language-conditioned 3D scenes	Interactive objects and spatial constraints	Physical environment diversity and grounding
Genie [9]	Learned visual dynamics	Latent, frame-level control	Complementary representation of world dynamics
LIGHT; SOTOPIA [12, 13]	Grounded fantasy play; social scenarios	Dialogue, actions, and social goals	Shared interaction and social evaluation
Generative Agents; Concordia [5, 6]	Persistent communities; grounded simulations	Memory and planning; Game Master adjudication	Direct precedents for memory and action grounding
AgentSociety; LMAgent [14, 15]	Societal and e-commerce environments	Large populations and domain-specific interactions	Scale and behavioral evaluation beyond this study
Agora	Premise-generated 2D social worlds	Typed shared state; repertoire and open proposals	Connects world authoring to inspectable social trajectories

Table 1. Research positioning by representation and mechanism. Entries describe the cited systems’ emphasis, not an exhaustive feature audit or a performance ranking. We do not treat unreported capabilities as absent.

unequal-budget architecture comparisons from model coverage audits and our explicit treatment of retrospective evidence.

The ODD protocol emphasizes explicit entities, scheduling, initialization, and model rationale for reproducible agent-based modeling [26]. We adopt these reporting priorities in Supplementary Section S1. Human-grounded agent simulation, such as models built from interviews and surveys and evaluated against the people they represent [27], sets a different evidentiary standard from fictional world coherence. Finally, work on complex contagion distinguishes information transmission from socially reinforced adoption [28]. Accordingly, the prospective intervention measures both warning diffusion and material treaty rescue; neither network density nor fluent conversation substitutes for coordinated action.

3 The Agora System

3.1 World Representation

Given a one-sentence premise p , the generation pipeline constructs

$$W = (E, G, R, K, A, M, P), \quad (2)$$

where E contains entities and their stable identities; G is traversable geometry; R contains physical, economic, and institutional rules; K specifies knowledge allocation and access; A is the initial action repertoire; M associates visual media with entities; and P records world identity and revision. Mutable state S_t contains locations, holdings, knowledge acquired during play, tasks, relationships, and event status. The schema is shared across worlds; its entities and rules depend on p .

This separation gives language-generated descriptions operational meaning. An archive must refer to an existing room, an item must have a valid owner, and a private fact must have a defined holder. Each visual asset is linked to the same entity identity used by the runtime. Changes then propagate through the state observed by subsequent actors rather than remaining only in dialogue.

3.2 Compositional Generation and Local Refinement

A planning stage identifies institutions, conflicts, resources, and activity loops. Specialized stages expand materials, rooms,

items, roles, main characters, knowledge, visual direction, wardrobe, property, and placement. The shared representation constrains references between their outputs. Visual generation produces room floors, character atlases, and objects under a common style specification; assembly connects these assets to the playable geometry and entity identities.

Validation targets four properties: semantic completeness, cross-modal identity, navigable and readable space, and persistent interactive state. Structural validators check references, inventories, asset provenance, and image contracts. Visual inspection and interactive checks supply different evidence about readability and use. When a check identifies an invalid character–item relation, that dependency can be revisited while unaffected stages remain fixed. This instantiates repair as a constrained part of generation, consistent with the broader PCG literature [2].

Running example. Tidal Embassy of Lost Languages turns linguistic scarcity into institutions and actions. Its generated configuration contains eight rooms, 25 named main characters, and 20 item types. Diplomats, smugglers, and memory divers receive different possessions and private knowledge. The setting exposes activities such as translating an archaic bribe, forging a linguistic seal, and rescuing a flooded treaty. These relations create reasons to move, exchange, withhold information, and seek cooperation. Supplementary Section S8 follows the full generation trace, including failed room images and localized visual repair.

3.3 Bounded Observation and Situated Intention

At round t , an activation policy selects agents. Agent i receives

$$o_i^t = (\ell_i^t, N_i^t, I_i^t, K_i^t, H_i^t, Q_i^t), \quad (3)$$

comprising location, nearby actors, inventory, accessible knowledge, episode and task memory, and selected directed relationships. Role and goals provide the decision context. The observation boundary is intended to preserve information asymmetry, an important distinction from omniscient social simulation [21].

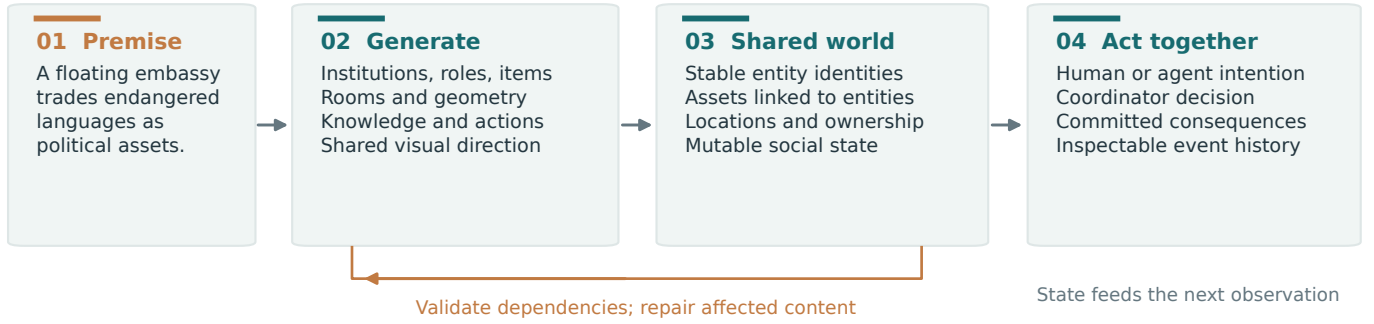


Figure 2. From premise to persistent consequences. Specialized generation stages populate a shared world representation. Validation connects semantic, visual, and spatial requirements; local repair revisits affected dependencies. Humans and agents act through the same world, while logged decisions and state changes support separate construction and interaction evaluations.

Effect	Required grounding
Speech	Addressable participant and a permitted channel
Transfer	Existing item, ownership, quantity, and co-presence where required
Trade	Transfer conditions and the partner’s acceptance
Creation	Declared available inputs and a resulting object identity
Status	Permitted target, bounded magnitude, and duration
Movement	Reachable destination and valid occupancy

Table 2. Examples of action conditions in the shared world. Linguistic plausibility alone does not establish that a requested effect is executable.

The agent chooses a partner and an intention, using either a generated action or an open proposal. Visible actions explain what an institution makes possible; proposals express intentions absent from that initial vocabulary. Open proposals in the audited implementation support four effect primitives: speech, transfer, item creation, and temporary status. Movement is a separate action route. These primitives can describe new contextual intentions, but they are not arbitrary executable programs or a general physics engine. Humans can approach an inhabitant, send messages, use or trade items, select a visible action, or formulate a proposal through the same world interface. This extends the shared grounded-play tradition of LIGHT [12] to premise-generated settings.

3.4 Coordinator-Governed State Changes

We write an open proposal as $q = (i, j, u, F, C)$: actor i , optional target j , language intention u , requested effects F , and declared consent evidence C . The model supplies a structured proposal; the audited coordinator core uses deterministic schema and state checks to return a decision and explanation,

$$\Gamma(W, S_t, q) = (d, r), \quad d \in \{\text{approve, revise, reject}\}. \quad (4)$$

Following the action-grounding role of Concordia’s Game Master [6], interpretation is separated from committing effects. An approved effect is applied by the state-transition machinery. Revision or rejection leaves the requested material effects unapplied, while the attempt can remain in the interaction history.

The resulting loop combines generative interpretation with

state constraints (Table 2). Ownership, quantity, location, and duration are represented explicitly. Social interpretation, including the meaning of an exchange or an accusation, remains model-mediated upstream. Declared partner acceptance is an input to validation; it is not itself proof of independently obtained human consent. Natural-language institutions are not automatically formalized as executable invariants by these checks. Consequently, the coordinator is a practical adjudication mechanism, not a proof of correct interpretation or universally valid social norms. Our evaluation reports observed outcomes rather than claiming exhaustive rule compliance.

4 Evaluation Design

We evaluate three links in the proposed mechanism: generated entities must satisfy shared representation contracts; inhabitants must be able to use both discoverable actions and contextual proposals; approved effects must persist while invalid effects are blocked. We separate generated worlds, interaction events, action results, and participant judgments as different units of evidence. A world can pass structural checks without producing engaging play. An event can contain several effects or attempts, so the number of logged events need not equal the number of adjudicated results. Longer trajectories describe the simulated population, while human judgments require a separate instrument and participants.

The evaluation addresses three mechanism questions: **RQ1**, how do shared contracts and local repair connect generated components into an executable world; **RQ2**, how are generated action opportunities and contextual proposals exercised in that world; and **RQ3**, how do coordinator decisions become persistent state changes, and what happens to invalid effects? Construction checks address RQ1, the five-world interaction study addresses RQ2, and execution audits with replayable traces address RQ3. The completed visual study characterizes the resulting artifacts. Internal generation controls and supplementary planner ablations report operating conditions and implementation boundaries. Table 3 states the unit and interpretation of each study.

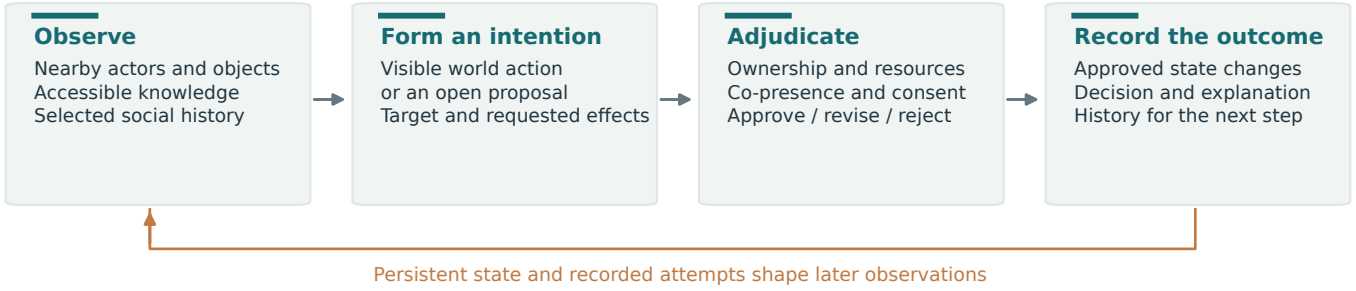


Figure 3. One interaction cycle. Bounded observations inform a visible action or an open proposal. The coordinator evaluates the proposed effects; approved changes update shared state. Both outcomes and failed attempts provide history for later decisions.

Study	Unit and scope	Evidence reported	Interpretation
Internal generation comparison	10 paired premises; 20 trials per method	Completion, structural quality proxies, cost	Descriptive reliability–cost tradeoff
Validation and repair	58 injected faults with clean checks; 5 repair cases	Detection, attribution, local repair cost	Specified contracts and seeded defects
Multiworld interaction	5 worlds; 2 rounds each; 53 events	Action pathways, dyads, outcomes, state changes	Initial behavior under common runtime settings
Cross-model audit	24 generations; 21 scripted 24-round runs	Completion, executed effects, rejected actions	Retrospective coverage and execution stress test
Historical pilot	3 prior runs; 1,052 events	Repeated ties and relational trajectories	Exploratory, confounded across worlds
Human model comparison	10 eligible returns; 40 image comparisons; 7 configurations	Exploration, map readability, character fit	Fixed pipeline and premise; sparse artifact preferences
Internal procedure control	20 structure comparisons; 10 requests	Same-system generation variants; full counts in supplement	Unequal budgets; no external-system comparison
Supplementary planner diagnostics	18 trajectories; 2 fixed worlds; 3 configurations	Input masking, available effects, and state replay	One centralized planner; implementation-specific counters

Table 3. Evaluation scope and evidence status. The studies distinguish construction, artifact preference, and executed consequences; evidence from different protocols is not pooled.

5 Construction Reliability and Internal Procedure Control

5.1 Paired Protocol and Completion

The generation comparison is an internal procedure control using the same Agora world representation. It uses Gemini 3 Flash Preview at temperature 0.2 and low thinking level for both methods; the single-pass condition is not an independently developed world-generation system. Five primary premises have three repetitions each; five held-out briefs have one each, giving 20 trials per method. The held-out set and evaluator were frozen before held-out generation. The requests include multi-sentence design constraints; this is not a strict one-sentence-input test (Supplementary Section S2). The analysis unit is the shared brief: primary-world binary outcomes use a majority over the three repetitions, while held-out outcomes use the single trial.

The single-pass baseline receives a 24,576-token output cap, three times the 8,192-token cap of one specialized stage. This provides a strong single-call allowance, but does not equalize total computation across methods. The compositional pipeline makes multiple calls and can use local retries. We therefore report first-pass and eventual completion separately, retaining failed outputs. The comparison evaluates the two com-

World-level result	Compositional	Single-pass
First-pass completion	9/10	5/10
Eventual completion	10/10	7/10
Primary mean tokens/trial	48,527	22,250
Held-out mean tokens/trial	77,556	38,922
Primary median latency (s)	101.2	55.8
Held-out median latency (s)	161.1	86.1

Table 4. Completion is aggregated at the premise level; cost and latency are trial-level summaries within each split. The primary split has 15 trials per method and the held-out split has five.

plete generation procedures; it does not isolate decomposition at equal total token budget. This joint reporting of reliability and cost follows the evaluation concerns in Kapoor et al. [25].

Compositional generation completes 10/10 worlds versus 7/10 for the baseline; first-pass completion is 9/10 versus 5/10 (Table 4). The paired eventual outcomes contain three compositional-only successes and no reverse discordances; the one-sided exact paired test gives $p = 0.125$. For first-pass completion there are five compositional-only and one baseline-only successes ($p = 0.1094$). These small samples support a descriptive improvement, not a statistically established gen-

eral advantage. Median generation is approximately 1.8–1.9 times faster for the single-pass method, which uses about half as many tokens.

5.2 Quality Proxies, Validation, and Repair

The six frozen quality measures show a mixed profile. Compositional outputs have higher mean merchant-inventory validity, premise-item recall, role uniqueness, room-canon adherence, and wardrobe-canon term coverage. The single-pass baseline has higher premise-agent recall (0.5593 versus 0.4632). The wardrobe proxy favors composition in nine of ten worlds (one-sided sign test $p = 0.0107$, unadjusted across the six measures). These are structural or lexical proxies; a higher score does not establish aesthetic preference or better play. We show all six in Figure 4, including the unfavorable comparison.

The seeded-fault suite mutates copies of five completed artifacts and invokes the production *deterministic validators*, without language-model or image-generation calls. It detects and attributes all 58 injected faults, with no flags in 58 corresponding clean checks. Faults cover missing semantic records, inventories, provenance mismatches, and image/assembly contracts. The clean checks reuse artifacts across categories, so they are not 58 independent worlds. This result concerns the specified fault suite; it is not 58 successful aesthetic judgments or evidence of universal defect detection.

For five injected role–inventory inconsistencies, localized refinement restores all five cases while preserving five unaffected intermediate stages. Mean model-call and token reductions relative to the full-rebuild reference are 48.0% and 39.9%. These observations support repair locality on the seeded failure type. A wider sample of naturally occurring defects is needed to estimate its general benefit.

5.3 Visual and Human-Evaluation Evidence

The five demonstration worlds connect generated maps, characters, objects, and persistent interaction. The Tidal Embassy trace includes rejected room images, local refinement, directional character atlases, and a 25-character interactive trial with movement, messaging, item use, trade, and a model-mediated reply (Supplementary Section S8). This is implementation and case evidence, not a blinded comparison of visual quality.

The completed visual questionnaire evaluates preserved maps, character samples, and standardized world-structure diagrams. Section 6 reports the seven-configuration model comparisons. The separately collected structure judgments compare internal generation procedures and are retained in Supplementary Section S5.1. Neither arm measures actual play experience or cross-system superiority.

6 Human Model Comparisons Within Agora

6.1 Comparison Target and Included Responses

The human model comparison holds the Agora pipeline and promise-city premise fixed while varying the recorded language-model configuration. Participants see a preserved map and the same sampling of two character images per side, under masked A/B labels. The image generator is shared. Thus the measured objects are the resulting visual world artifacts under each configuration, not stand-alone LLM image generation

and not competing world-authoring systems.

Ten of eleven returned English visual packets meet the released eligibility rule. V04 contains all eighteen choices but lacks cover confirmations and remains pending. The ten eligible returns supply forty image comparisons, covering all 21 pairs among seven configurations, and 120 item responses. Four pairs have one rater, sixteen have two, and one has four. Each comparison asks which world the respondent would rather explore, which map is easier to read, and which characters better fit their world. These criteria remain separate; they are not components of a validated overall-quality scale. The protocol and complete pairwise counts appear in Supplementary Section S5.

6.2 Observed Votes by Model Configuration

Table 5 adds a descriptive view of the model comparisons. For each configuration and question, we pool its observed wins and half-weighted ties and divide by its judgeable comparisons. Not sure choices are excluded from that denominator and remain in the full counts. This roll-up was added after responses were available to make the pairwise results readable; it is not a preregistered ranking. Opponent weights differ because the allocation and one cover exclusion leave unequal pair coverage.

Gemini 2.5 Flash receives eight of ten exploration preferences (80.0%) and seven of nine judgeable character-fit preferences (77.8%). GPT-5.6 Terra receives nine of thirteen exploration preferences (69.2%) and eleven of thirteen map-readability preferences (84.6%); GPT-5.6 Sol receives nine of eleven readability preferences (81.8%). The contrast within Sol is substantial: only four of eleven comparisons favor it for exploration. Gemini 3.1 Flash Lite receives six of ten exploration preferences but no readability wins and one readability tie (0.5/10). These are observations about fixed artifacts, not statistically established model superiority.

6.3 Criterion-Specific Pairwise Outcomes

Figure 5 reports all exact model pairs for exploration, map readability, and character–world fit. Each model name denotes the recorded world-generation configuration; the pipeline uses a shared image generator, so the comparison does not isolate an LLM’s ability to render pixels. Only one prompt has matched archived images.

The three criteria give different outcomes. Both raters prefer the GPT-5.6 Sol artifact to Gemini 3.5 Flash Lite for map readability, but both prefer the latter for exploration and character fit. The pairwise matrix preserves these distinctions instead of collapsing them into a single model ranking. Across the image arm there is one readability tie, one character-fit tie, and two character-fit abstentions. Four pairs now have one rater, sixteen have two, and one has four. This sparse, single-scene sample supports inspection of these particular outputs, not a stable ordering of model families or general claims about visual quality.

Separate internal control and unmeasured structure quality. The questionnaire also collected twenty comparisons between two generation procedures inside Agora. Their full results remain in Supplementary Section S5.1; the observed diagram preferences favor the single-pass procedure in aggregate.

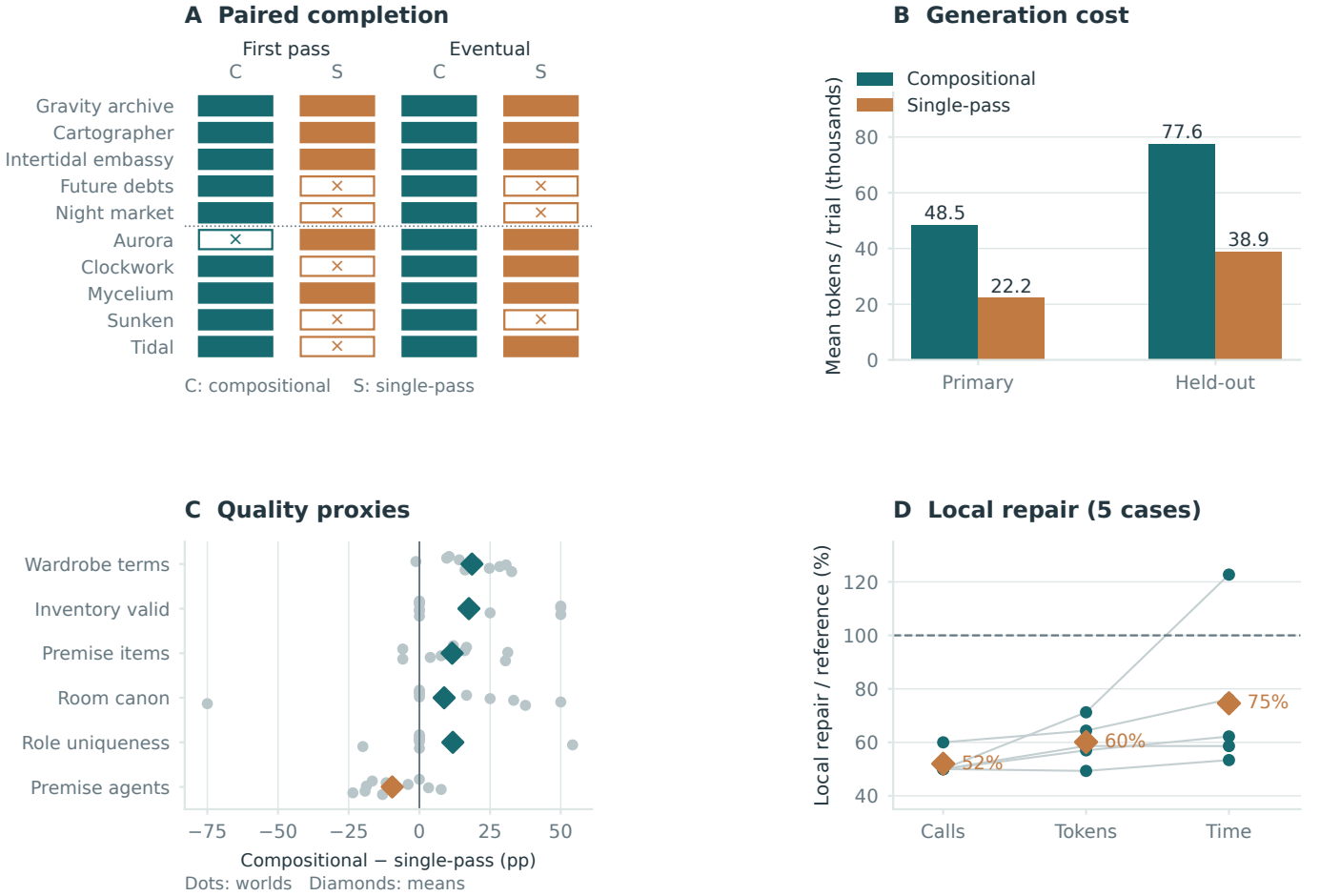


Figure 4. Construction evidence at its measured resolution. (A) Paired premise-level completion; the primary split uses majority outcomes. (B) Per-trial token use by split; Table 4 reports latency. (C) Differences in structural and lexical quality proxies, including the metric favoring the baseline. (D) Local-repair resource ratios for five cases. All values come from frozen records; no new model trials were run for this revision.

Recorded configuration	Explore	Map readability	Character fit
Gemini 2.5 Flash	8/10 (80.0%)	7/10 (70.0%)	7/9 (77.8%)
Gemini 3.1 Flash Lite	6/10 (60.0%)	0.5/10 (5.0%)	4/9 (44.4%)
Gemini 3.1 Pro Preview	2/11 (18.2%)	4.5/11 (40.9%)	3.5/10 (35.0%)
Gemini 3.5 Flash Lite	6/14 (42.9%)	2/14 (14.3%)	6.5/14 (46.4%)
Gemini 3.7 Flash	5/11 (45.5%)	6/11 (54.5%)	5/11 (45.5%)
GPT-5.6 Sol	4/11 (36.4%)	9/11 (81.8%)	5/11 (45.5%)
GPT-5.6 Terra	9/13 (69.2%)	11/13 (84.6%)	7/12 (58.3%)

Table 5. Observed model-comparison votes on the single promise-city scene. Each cell is (wins + half ties) / judgeable comparisons, followed by its percentage. Not sure is excluded; half-integer numerators indicate a tie. Rows retain recorded model identifiers in fixed order. Models have unequal opponent weights, so these descriptive scores are not a balanced tournament, a general capability ranking, or independent samples across rows.

Both conditions share Agora’s representation, so these controls do not compare Agora with an independent system. They also do not measure structural differences *between the seven model configurations*. Human model comparisons here cover maps and characters; cross-model world-structure preference and actual play experience were not measured.

7 Interaction Across Five Generated Worlds

7.1 Protocol and Metric Definitions

The five held-out worlds are Aurora Court of Migrating Cities, Clockwork Rain Conservatory, Mycelium Patent Bazaar, Sunken Satellite Monastery, and Tidal Embassy of Lost Languages. Each contains 24–25 main characters and 11–12 discoverable actions spanning five interaction families. At least five activities per world derive from its premise.

Each accepted run uses Gemini 2.5 Flash for two rounds,

Explore preference

G2.5 Flash		100 (n=2)	100 (n=1)	100 (n=2)	100 (n=1)	50 (n=2)	50 (n=2)
G3.1 Lite	0 (n=2)		50 (n=2)	100 (n=2)	100 (n=2)	100 (n=1)	0 (n=1)
G3.1 Pro	0 (n=1)	50 (n=2)		50 (n=2)	0 (n=2)	0 (n=2)	0 (n=2)
G3.5 Lite	0 (n=2)	0 (n=2)	50 (n=2)		50 (n=2)	100 (n=2)	50 (n=4)
G3.7 Flash	0 (n=1)	0 (n=2)	100 (n=2)	50 (n=2)		100 (n=2)	0 (n=2)
GPT Sol	50 (n=2)	0 (n=1)	100 (n=2)	0 (n=2)	0 (n=2)		50 (n=2)
GPT Terra	50 (n=2)	100 (n=1)	100 (n=2)	50 (n=4)	100 (n=2)	50 (n=2)	
	G2.5 Flash	G3.1 Lite	G3.1 Pro	G3.5 Lite	G3.7 Flash	GPT Sol	GPT Terra

Map readability

G2.5 Flash		100 (n=2)	0 (n=1)	100 (n=2)	100 (n=1)	50 (n=2)	50 (n=2)
G3.1 Lite	0 (n=2)		25 (n=2)	0 (n=2)	0 (n=2)	0 (n=1)	0 (n=1)
G3.1 Pro	100 (n=1)	75 (n=2)		100 (n=2)	0 (n=2)	0 (n=2)	0 (n=2)
G3.5 Lite	0 (n=2)	100 (n=2)	0 (n=2)		0 (n=2)	0 (n=2)	0 (n=4)
G3.7 Flash	0 (n=1)	100 (n=2)	100 (n=2)	100 (n=2)		0 (n=2)	0 (n=2)
GPT Sol	50 (n=2)	100 (n=1)	100 (n=2)	100 (n=2)	100 (n=2)		50 (n=2)
GPT Terra	50 (n=2)	100 (n=1)	100 (n=2)	100 (n=4)	100 (n=2)	50 (n=2)	
	G2.5 Flash	G3.1 Lite	G3.1 Pro	G3.5 Lite	G3.7 Flash	GPT Sol	GPT Terra

Character-world fit

G2.5 Flash		100 (n=1)	100 (n=1)	100 (n=2)	100 (n=1)	50 (n=2)	50 (n=2)
G3.1 Lite	0 (n=1)		50 (n=2)	50 (n=2)	50 (n=2)	100 (n=1)	0 (n=1)
G3.1 Pro	0 (n=1)	50 (n=2)		75 (n=2)	50 (n=2)	0 (n=2)	0 (n=1)
G3.5 Lite	0 (n=2)	50 (n=2)	25 (n=2)		50 (n=2)	100 (n=2)	50 (n=4)
G3.7 Flash	0 (n=1)	50 (n=2)	50 (n=2)	50 (n=2)		100 (n=2)	0 (n=2)
GPT Sol	50 (n=2)	0 (n=1)	100 (n=2)	0 (n=2)	0 (n=2)		100 (n=2)
GPT Terra	50 (n=2)	100 (n=1)	100 (n=1)	50 (n=4)	100 (n=2)	0 (n=2)	
	G2.5 Flash	G3.1 Lite	G3.1 Pro	G3.5 Lite	G3.7 Flash	GPT Sol	GPT Terra

How to read the matrices

Cell = preference for the row artifact against the column artifact.
 Same = half a vote; Not sure is excluded.
 n = judgeable votes for that pair/item.
 Mirrored cells reuse the same votes.

10 eligible returns | 40 image comparisons
 21 pairs | one shared promise-city scene
 Pair coverage: 4 x one, 16 x two, 1 x four
 Recorded identifiers; no global ranking.

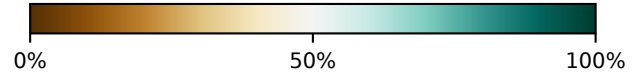


Figure 5. Images: all 21 pairs, with separate evaluation criteria. Cells report the percentage preferring the row artifact over the column artifact, counting ties as half votes; n is the number of judgeable votes. Not sure is outside the denominator. Mirrored cells describe the same comparisons, not additional observations. All outputs share one promise-city premise and the image pipeline; labels abbreviate archived backend identifiers. Exact counts, including abstentions, are in Supplementary Section S5.

activation probability 0.12, and a fixed world-specific seed (26080411–26080415). Agent-profile generation is not repeated during these runs. Image and video generation are disabled, and all five runs pass the world-identity audit. The runtime settings are shared, but the seeds and world contents differ: this is a descriptive multiworld study, not a paired intervention isolating premise effects.

An *event* is a recorded interaction. An observed participant is any actor or target ID, including the human interactor where present. A dyad is an unordered pair of distinct actor and target IDs. If n_{ij} is its event count, the repeat fraction is

$$\rho = \frac{\sum_{\{i,j\}} \max(n_{ij} - 1, 0)}{N_{\text{events}}}. \quad (5)$$

Action entropy is $H = -\sum_a p(a) \log_2 p(a)$ over logged route identifiers. An open event has a `proposed_route`; an explicit world-action event has a `world_action_route`. Other routes can still encode premise-specific activities. Thus these categories identify execution pathways, not an exhaustive semantic classification of world relevance.

Action-result success is computed separately from round summaries:

$$s = \frac{\sum_{w,t} n_{w,t}^{\text{success}}}{\sum_{w,t} n_{w,t}^{\text{results}}} = \frac{49}{56} = 87.5\%. \quad (6)$$

An event can include multiple effects or attempts. Neither 53 events nor 37 dyads is the denominator of this rate. Supple-

mentary Section S1 records the per-world numerators and denominators.

7.2 Action Pathways and Consequences

The runs produce 53 events, 37 within-world unique dyads, ten approved open proposals, and nine explicit world-action events (Table 6). Sunken Monastery uses eight explicit world actions, including purging corrupt packets and negotiating bandwidth tithes. Clockwork instead produces seven open-proposal events out of twelve. This contrast shows two available execution pathways in use, without proving that one causes better behavior.

In Clockwork, a jammed weather-sheet press becomes a local investigation. Marius confronts Valerius about the failure; Seraphina asks about etching; Valerius discusses sheet authenticity. In round two, the same participants revisit sabotage, inspect a brass folio, and redirect suspicion. Six actions form this three-character sequence; a seventh open proposal concerns the market impact of the jam. The approved effects update directed relationships while the press failure remains part of shared context. The complete episode diagram is retained in Supplementary Section S7.

The resulting state changes differ across worlds. Clockwork accumulates +3 net trust and +21 influence/fear, whereas Aurora and Tidal accumulate +70 and +88 trust with zero net influence/fear. These are simulated state variables produced through the model-mediated interaction machinery. They show that the traces have persistent consequences, but they are not independent psychological measurements or evidence that the premise alone caused the differences.

7.3 Human Reachability and Failed Attempts

Thirteen events target the human interactor across four worlds; Clockwork has none. The logs establish that agents can select the player as a partner within the shared world. They do not establish how often a human replied, whether cooperation succeeded, or whether participants enjoyed the encounters. Those questions require participant-level outcome measures.

Seven of the 56 adjudicated action results are unsuccessful. Retaining these attempts in the history permits later revision, but this study does not measure whether failure feedback improves subsequent success. The evidence supports the availability of an outcome-sensitive interaction loop, consistent with the broader importance of environment feedback in language agents [17–19].

8 Persistent State and Execution Evidence

8.1 A Separate One-Sentence Corpus

The August 25 archive adds eight recorded model identifiers and three shared one-sentence prompts: a city taxing spoken promises by weight, a railway greenhouse allocating access through crop-care obligations, and a courthouse rebuilding a shoreline from confiscated objects. This later pipeline requests 12 agents and four player slots. We re-audit its 24 generation outcomes separately from the earlier ten-brief comparison: the prompt set, implementation, and available models changed, so their scores are not pooled.

Seventeen of 24 trials yield complete artifacts, with nine

first-pass completions. Five models complete all three prompts; Gemini 3.5 Flash Lite and GPT-5.6 Terra each complete only the first, and GPT-5.6 Luna completes none. Supplementary Table S2 retains every failure. These are recorded API identifiers, not guarantees of immutable provider snapshots. One generation per model–prompt cell supports a coverage diagnosis, not a stable model ranking. Although the archive calls the suite hidden/confirmatory, it lacks a pre-execution immutable registration, and some scoring and retry policies were revised after development runs. We treat it as retrospective evidence.

8.2 What the Longer Harness Actually Measures

Seven first-prompt artifacts support three repeated 24-round trajectories each. The harness makes six calls per trajectory, at rounds 1, 5, 9, 13, 17, and 21; each call plans 12 actions over four rounds. Its planner observes summaries of *all* agents and recent events. It is a centralized multi-character policy, distinct from the locally observed agents in Section 3. Scripted requests, an invalid transfer, a resource shock, and a cross-room request probe execution. The prompt explicitly asks for thread continuation, legal recovery, and human follow-through; these behaviors cannot be claimed as spontaneous emergence or participant engagement.

The 21 trajectories contain 1,512 attempted model actions, of which 1,365 succeed (90.3%). Among successful actions, 398 (29.2%) record an inventory or location mutation. Among 314 approved proposals, 150 (47.8%) declare at least one non-speech effect; status-only effects count in this latter measure. These denominators address different questions and are not interchangeable. Figure 7 reports model-specific counts and repeat ranges instead of a weighted quality score.

8.3 Failure Modes and Evidence Boundaries

Gemini 3.7 Flash succeeds on all 216 attempts, but only 19 successes change inventory or location and only four of 22 approved proposals have non-speech effects. Terra’s 18 approved proposals are all speech-only, despite 214/216 successful actions. Conversely, Gemini 2.5 Flash has 42 failed attempts but 34/45 approved proposals with non-speech effects. These observations motivate reporting *what changed* alongside whether an action executed, consistent with analytical trajectory evaluation [24].

All 147 failed model actions and all 21 scripted invalid transfers leave their recorded state hashes unchanged. Final-state serialization round trips match in 21/21 runs; this checks serialization, not recovery after a process crash. Of 1,021 recorded relationship updates, 1,013 have the same $(\Delta\text{trust}, \Delta\text{affection}, \Delta\text{fear}) = (4, 2, 0)$. The custom-action handler supplies exactly this default for non-aggressive actions. Thus rising relationship values in this harness largely reflect execution rules rather than independently demonstrated social development. Supplementary Section S2 provides the full audit and replication boundaries.

8.4 From a Proposal to a Later Observation

A retained Clockwork trajectory makes the state-transition mechanism concrete. At round 11, Thistle Thorne proposes giving one Corrosive Inkwell to Elara Moss. The structured

World	Events	IDs	Dyads	Repeat	H (bits)	Open	World route	Human-targeted
Aurora Court	9	12	7	22.2%	1.447	1	1	2
Clockwork Conservatory	12	8	6	50.0%	3.252	7	0	0
Mycelium Bazaar	9	12	7	22.2%	1.392	0	0	4
Sunken Monastery	11	13	9	18.2%	1.823	1	8	3
Tidal Embassy	12	8	8	33.3%	1.208	1	0	4
Total	53	–	37	–	–	10	9	13

Table 6. Two-round interaction records. IDs count observed participants, including the human where applicable. Open and world-route counts follow logged route prefixes; other routes may also be premise-specific. Dyads are unique within each world.

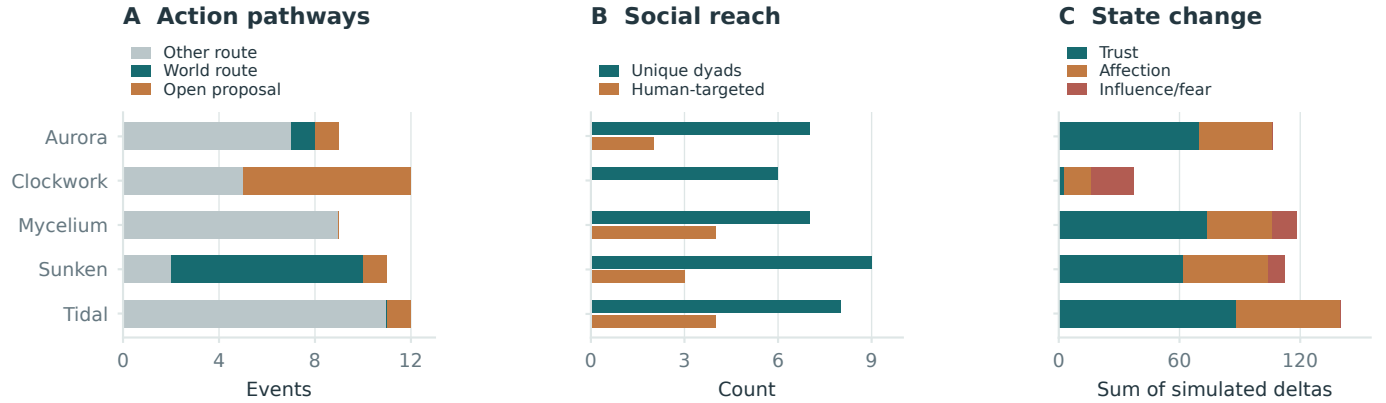


Figure 6. Interaction profiles under common runtime settings. (A) Logged action pathways. (B) Unique dyads and events targeting the human interactor. (C) Sums of model-mediated relationship changes. The five worlds have one short run each; bars are descriptive totals, not population estimates.

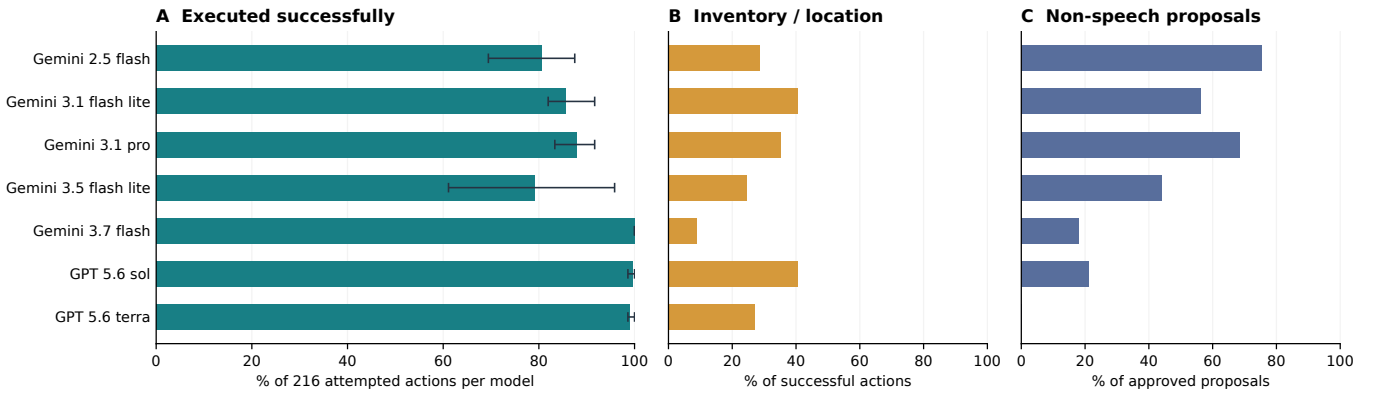


Figure 7. Successful execution does not ensure rich consequences. Seven generated worlds, each with three 24-round centralized-policy runs. (A) Success per attempted action; whiskers are the minimum and maximum over three runs, not confidence intervals. (B) Inventory/location mutations among successful actions. (C) Non-speech effects among approved proposals, including status changes. All comparisons couple a model with its own generated world; they do not isolate policy capability on identical environments.

effect names the existing item, source, recipient, and quantity. The coordinator checks ownership, co-location, and effect bounds, then records a transfer. In the next planning observation at round 13, the item appears in Elara’s inventory and is absent from Thistle’s. A separate scripted attempt to transfer an unowned item is rejected with identical before/after state hashes. This traces the connection from a generated possession through a proposed action to a persistent consequence and a later observation; the transfer alone does not establish completion of its stated goal. Supplementary Section S3 identifies the source records and reports all 18 planner-diagnostic trajec-

tories, including the full input-history and action-space ablation tables.

9 Historical Context

Three earlier runs add 1,052 events across an adventuring guild, a black-market auction, and a storm-struck cruise. Between 23.4% and 36.5% of events revisit an existing dyad, while the largest connected components contain only 11–13 participants. These records motivate studying communication across local clusters, but configurations differ and spatial scheduling can produce the observed structure. Full tables and trajectories ap-

pear in Supplementary Section S4; they are not pooled with the later scripted harness.

10 Discussion and Limitations

Connecting authoring constraints to runtime consequences. Agora’s mechanism uses the same generated entity identities and relations through world assembly, action grounding, and state updates. A possession created during authoring becomes an ownership constraint at execution time; a committed transfer changes what later actors observe. Dependency-directed repair maintains the corresponding construction contracts. Language-driven game construction [3, 4], shared human-agent play [12], and Game Master adjudication [6] provide direct precedents. Our contribution is the coupling of these stages through a premise-conditioned representation and inspectable state transitions. The construction and execution records show how that coupling operates.

Discoverability and expressiveness are separate design needs. A generated repertoire explains what inhabitants can do in a new setting. Open proposals let a particular encounter depart from that vocabulary. The Sunken and Clockwork traces illustrate both pathways. A proposal specifies contextual intent through bounded effects, and the coordinator checks those effects against shared state. The transferred-item example exposes the full path from intent to a later inventory observation. Supplementary diagnostics retain the counts under restricted action and history inputs; differences in available handlers must be considered when interpreting those counts. Useful human collaboration requires evidence from actual play.

Internal procedure controls have limited scope. Ten shared briefs compare staged and single-pass generation under the same representation, with different total computation and retry opportunities. The observed completion gains characterize that implementation comparison, not the value of Agora’s shared representation or its superiority to an external system. The supplementary diagram judgments favor single-pass outputs in aggregate and remain reported in full. Both favorable and unfavorable control outcomes have this same restricted scope.

The visual study is narrow and exploratory. Ten eligible coded returns supply forty model-image comparisons; a separate arm supplies twenty internal structure controls. One of eleven returns lacks cover confirmations. The image arm covers one scene, with one to four ratings per exact pair, and the structure arm shows containment diagrams rather than a complete interactive world. The source identities are masked, but the artifacts share a generation pipeline and retain unequal compute. The organizer reports unpaid online volunteer recruitment; an ethics approval or exemption record and completion times remain undocumented. Accordingly, the findings characterize these returned artifact judgments, not a stable model ranking or human enjoyment.

Execution is not behavioral validation. The five locally observed runs last only two rounds with one seed each. The 21 longer runs use centralized block planning, scripted human events, and each model’s own generated world. The additional planner ablations control the initial world and model but retain

centralized decisions and scripted interventions. Their high execution success does not validate emergent coordination; identical relationship increments expose a contribution from default state-update rules. Historical worlds are confounded by prior configurations. Relation deltas are model-mediated state variables, and route entropy depends on the action vocabulary. Neither measures human trust, fear, affection, or social intelligence directly. LLM judging also carries systematic biases [23]; behavioral fidelity would require independent human data and explicit validation targets, as in human-grounded agent studies [27].

Reproducibility and human participation. Hosted models may change despite a recorded model name; available settings, seeds, exclusions, and frozen evidence are documented in Supplementary Section S1. The present human study is complete at its reported scope. Evaluating refusal, outcome legibility, and collaboration during play is a separate future study. The prospective local-memory protocol in Supplementary Section S6 is also future work, not evidence used for the paper’s conclusions. Fictional roles do not validate demographic models.

11 Conclusion

Agora connects a short premise to a persistent social world through three mechanisms: a shared representation with dependency-directed repair, a situated interface combining generated actions and open proposals, and coordinator-governed state transitions. Generated possessions, locations, and identities become constraints on action; approved effects alter the state available to subsequent decisions. Construction checks, action traces, and deterministic replay make these connections inspectable. The completed visual study characterizes the rendered artifacts, while supplementary diagnostics document the tested implementation’s scope. This provides a concrete foundation for studying consequential human-AI interaction in generated settings. Collaboration during play and decentralized social behavior remain directions for subsequent evaluation.

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